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SPECIAL SECTION

Selected contributions to the "Workshop on Design and Implementation of Data Analysis Systems," held at Universität Heidelberg March 23–26, 1995.

Interactive Graphics for Data Sets With Missing Values—MANET

Antony UNWIN, George HAWKINS, Heike HOFMANN, and Bernd SIEGL

Missing values are a problem for statistical methods. This applies just as much to modern methods such as interactive graphics as to more classical methods. The MANET software has been developed for keeping track of missing values in interactive graphics analyses and for investigating new interactive graphics tools.

Key Words: Interactive graphics; Missing values.

1. INTRODUCTION

Missing values are a problem. They arise in surveys where people fail to answer questions or refuse to, they arise in observational studies through gaps in recording, they even arise in carefully designed sets of experiments when particular cases fail. Missing values may also be structural, people who do not drink whiskey cannot tell you what they like about the taste.

Time series may have many different kinds of missings too: structural missings (there are no closing share prices on Sundays), missings whose values could be found with additional effort (sales figures for a particular period could be calculated from stock changes or by some other indirect way if the actual figure is not available); missings where values existed but cannot be recovered afterwards (if a patient's temperature was not taken yesterday we cannot later find out what it was).

Geographical data may have structural missings (e.g., much data for the new Bundesländer in Germany is not available in comparable form for the years prior to unification) as well as other kinds. Geological data sets may include locations at which not all variables were measured. In ornithological data sets, sites at which no birds have been

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seen may be treated as missings rather than as zeros; the same applies to records of fish catches. In both these examples the use of missing rather than zero would have serious consequences for statistical models but can be advantageous for interactive graphical analyses in displaying patterns of missings across large numbers of species.

When there are few missings they may often be ignored or imputed (for instance using expected values or "hotdecking"), but when there are many missing values this is unwise. In the discussion of political opinion polls it is common to mention the percentage of "don't knows" but then to ignore them in reporting the percentages supporting the various parties. This has the effect of imputing the same proportions for party support to the don't knows as to those who did express an opinion.

It is very much the user's responsibility to decide what to do about missing values. Apart from ignoring or imputing them, it is possible to try predicting them using whatever other variables or information are available. No method is optimal in all situations and the philosophy behind MANET is that the user should be kept aware that the data are incomplete so that they qualify properly any conclusions they reach, whether they reached them by dropping the missings or by predicting them in some way. Vach (1994) gave a useful summary of the current status of statistical theory and missing values.

Interactive graphics are a valuable new tool for exploring data and there is an excellent implementation of these ideas in Data Desk (Vellemann 1992). But until now interactive graphics have, like many other statistical tools, largely ignored missing values. This is dangerous because false impressions may be created that are difficult to dislodge. Surveys of tourists ask questions about expenditure on holiday, often in surprising detail. Not only is the quality of the responses obtained questionable but many respondents give no response at all. Studying scatterplots of amount spent against length of holiday, number in group, or other factors may imply conclusions that are based on unrepresentative subsets of the respondents.

1.1 EARLIER WORK

Diamond Fast (Unwin and Wills 1988) is a package written specifically for exploring multiple time series with interactive graphics. It represents time series either by points or by line sections drawn between the points or by both. Similar to how any irregular or regular pattern of times may be displayed, we could consider the representation of a series by points alone as displaying the missings with blanks and the representation with line sections as imputing linear interpolations to missing values (of course, we can only talk about missings in series where values might actually have existed at the intermediate time points). If a series is recorded only at regular time points (such as each year) then missing values arise if there is no data for particular years. Figure 1 shows the batting average of Joe DiMaggio, the famous baseball star, over his career. It is apparent that there are no values for the years he was in the armed services during the war.

If there are several irregular series being considered then we may restrict the term missing value to cases where values are recorded in some series at time t but not in all. It was intended to implement a variety of missing value representations in Diamond Fast (such as using a special symbol for missings or not plotting a line segment across missings while using linear interpolation elsewhere) but the display possibilities already

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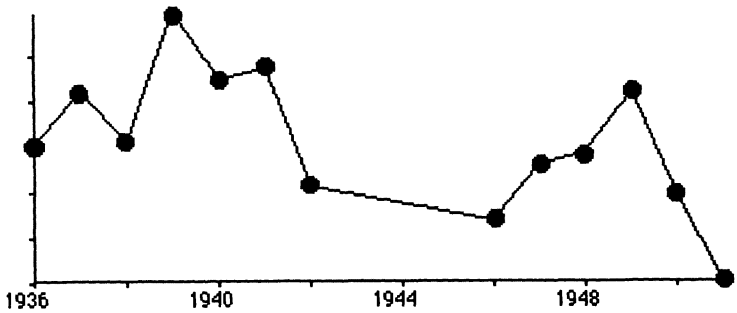

Figure 1. Joe DiMaggio's batting average from 1936 to 1951.

available worked well except in series with very large numbers of points and they require special treatment anyway. The one rule that was very clear was that using 0 or 9,999 or any other numeric substitution led to very ugly and unhelpful plots.

REGARD (Unwin 1994) was written as a tool for exploring spatial data. In many applications—geological and geographical, for example—values were frequently missing and it was necessary to keep carefully in mind which data were available and where. To this end a missing value plot was developed that displayed two horizontal bars for each variable. Figure 2 refers to three variables recorded for regions of the European Community in 1987 and 1990, although the map includes the regions of the EC as they were in 1991.

For each variable the bar on the left represents the cases that were not missing and the bar on the right represents the cases that were missing. The usual linking capabilities apply and currently all cases for which unemployment data are missing have been selected. The corresponding points in the other bars have been highlighted so we can see that the areas without unemployment data are mostly also without the other variables. The short sections highlighted among the nonmissings in the other two variables show that there is at least one region with values there but not for unemployment. The map suggests that this is probably the Algarve in Portugal, which can readily be confirmed by linking from the map. The missing value plot proved both useful not only for its original purpose of simply displaying the locations with missing values but also for inspecting patterns of missings using REGARD's interactive querying tools. It highlighted how important a proper treatment of missings is.

1.2 EXAM MARKS: AN ILLUSTRATIVE DATA SET

To illustrate the problems of missing values in some detail consider the data set of marks from a recent statistics exam. One hundred ten students had to answer 10 multiple choice questions and then five out of a possible eight longer questions.

Performance on pairs of questions could be compared either by linking from the histogram of marks on one question to the histogram of marks on another or by drawing a scatterplot. Neither is satisfactory as no pair of questions was answered by all students. The two histograms will include all the marks for each individual question but linking will only apply to cases where students answered both questions, making interpretation of

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Figure 2. Gaps in some European Community data 1987.

the highlighted histograms difficult. On the other hand, the scatterplot will only include students who answered both questions, so the distributions of marks of the individual questions are not fully represented.

Distributions of marks across all eight main questions could ideally be compared by a set of boxplots. These will be based on different numbers of students (which may not be of much importance in this example unless one question was attempted by very few). If they are drawn together, then a possible software default (as, for instance, in Data Desk) is to base the plots only on cases that exist for all variables resulting in eight blank plots, since no student answered all eight questions. If the boxplots are drawn separately, then they may have to be individually scaled to make them comparable (since software scaling defaults are always based on the maximum and minimum of the variable, a question which was badly answered would have a low maximum and the plot of the marks would be differently scaled than the plot for a question on which marks ranged from 0 to full marks).

In both these cases it is important to be aware of which cases are missing and to keep track of them. Some packages do give numbers of missings associated with a plot (for instance, BMDP New System for Windows) but this does not suffice for linking across windows and is not fully informative in scatterplots where missings may be missing on both variables or just on one or the other.

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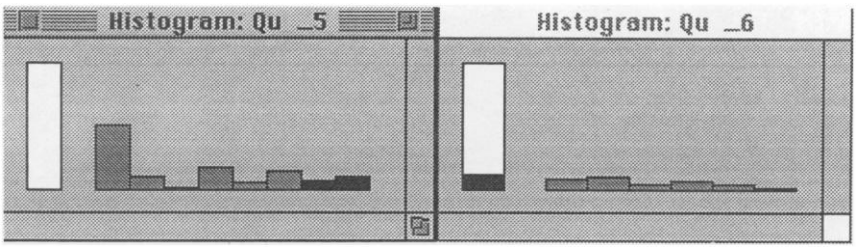

Figure 3. Histograms of Marks for Questions 5 and 6. The white columns to the left represent those students who did not answer the question.

1.3 MISSING VALUES IN MANET

MANET (Missings Are Now Equally Treated) has been developed to extend interactive graphics tools to deal with missing values. Each plot requires a different treatment and so this has involved implementing some obvious simple extensions while also developing new ideas to cope with specific problems. The common aims have been to ensure that the user is always aware of missing values, that no cases are ignored because they do not have complete data for all variables and that, when linking, all cases are treated equally as far as possible, whether they are complete or not.

Histograms are extended by introducing a white missing value bar to the left of the main plot. Its height is proportional to the number of missings and highlighting works just as with the other histogram classes. Figure 3 shows the histograms for Questions 5 and 6 with the students who did best on Question 5 selected (none of them answered Question 6).

Note that these plots have been scaled by default individually so that they are not directly comparable. There were 50 points available for each question and the raw results showed that only a limited number of possible marks were actually used by the examiners. An ideal display would choose a common scale for the two histograms which captured this aspect of the data. It is easy to do this in MANET, where the philosophy is that the user should look at many plots and be able to do so quickly and flexibly. However, for the purpose of showing that those who did best on Qu 5 did not answer Qu 6, the default plots are sufficient.

Scatterplots require special treatment and this is shown by comparing scatterplots of the marks for Questions 5 and 6 (Fig. 4). The left plot is from Data Desk and ignores students who did not answer both questions, that is it only shows 15 out of 110 possible points. (In some cases more than one student obtained the same pair of marks so that a point may represent more than one student.) The plot on the right is from MANET and shows the same 15 students but also the 38 students who answered Question 5 but not Question 6 along the special x axis at the appropriate positions (i.e., at their x values) and similarly the 15 students who answered Question 6 but not Question 5 along the special y axis. Additionally, the 42 students who answered neither question are represented in a special box underneath the plot. These numbers can be obtained by interrogating the plot. To ease identification of the missings there are further boxes for those cases that have neither x nor y values. The origin in the MANET plot is automatically chosen so

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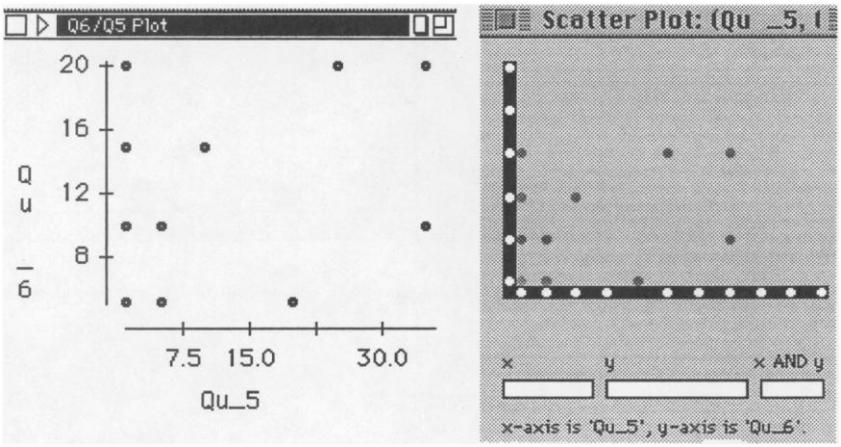

Figure 4. Scatterplots of Marks of Students on Questions 5 and 6. Plot from Data Desk on the left, from MANET on the right.

that the axes are drawn to the left and below all data with both x and y values. When more than one student is represented by one point, MANET plots the point more brightly. Brightness scaling for overlapping of points can be set in the plot's parameters box.

The main point to make in comparing these two plots is that the scales were automatically generated within the respective programs so that the MANET plot adds not only additional information about the existence of missings but also shows that the best results on both questions were achieved by students who only answered one of the two. Exploratory data analysis involves getting as much information out of a data set as possible and this is an example where MANET extracts more than packages that do not represent missings. Of course, the user's judgement is still very much necessary, both to check the information, perhaps by a further study of the raw data, and to decide whether it is important enough to report.

There is an option for scatterplots (as for all plots) to display without missing values. One view can emphasize aspects of the data set which another does not, but usually it is best to look at both.

Displaying missing values with boxplots is harder and the solution adopted in MANET is to automatically generate a missing value plot including all variables for which boxplots have been requested. This is very similar to the REGARD missing value plot. A single horizontal bar is drawn for each variable. If there are missings then a line is drawn separating the bar into a section on the right to represent the missings while the rest represents the cases for which values are available. The missing values plot is extremely effective in its own right for comparing patterns of missings across a large number of variables, but as it is drawn in a separate window it is not forcibly tied to the boxplots.

Bar charts handle missing values in the obvious way of allocating a special category to them, but as with histograms the missing value bar is white rather than shaded to clearly differentiate it from the others. Currently, the missing value bar is drawn as the

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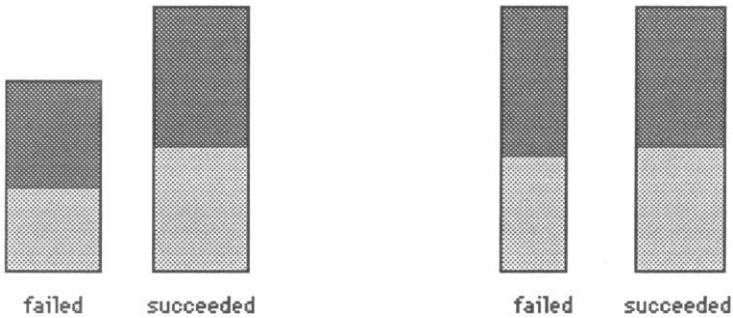

Figure 5. Bar Charts of Numbers Passing and Failing, Proportional to Height on the Left and to Width on the Right. Numbers answering Qu 5 are highlighted.

rightmost bar, partly because it is often coded as the last category and partly to clearly differentiate barcharts from histograms. As it is intended to introduce an interactive capability to reorder categories in bar charts, the default positioning of the missing value bar will be of less importance.

1.4 TOOLS FOR CATEGORICAL DATA IN MANET

Although the main aim in developing MANET has been to treat missing values properly, the opportunity of introducing new interactive tools that are not available elsewhere has also been taken. The main weakness in linking to bar charts is that it is difficult to judge the proportions highlighted. Selecting those who have done well on Question 5 and comparing what proportions have passed or failed out of all those who passed and failed is difficult (see Fig. 5, left side), even though it is possible to interrogate both bars to get the actual numbers. To solve this problem, a proportionate version of the display has been provided. If the cursor is moved to the foot of the window, it changes to a vertical double-headed arrow. Clicking changes the bar chart to one in which all bars have the same height but their width is now proportional to the numbers in the category (see Fig. 5, right side). This makes comparisons of highlighted proportions very easy by eye. If the mouse is now at the foot of the window, the cursor becomes a horizontal double-headed arrow and one click returns the original bar chart. Further graphical extensions displaying confidence intervals have been suggested by Hummel (1994).

The proportionate version of the bar chart is in fact a one-dimensional mosaic plot (see, e.g., Friendly 1994) but with linking. It was a natural extension to consider implementing a full mosaic plot and MANET can handle up to 8 binary variables or 256 categories in all. Each possible combination of categories is a cell and is represented by a rectangle proportional in area to the number of cases in it. The resulting display is a mixture of rectangles of various sizes and, commonly, a large number of empty cells that are marked by points or lines with a zero through them. A mosaic plot shows quickly whether particular combinations or clusters of cases dominate the data set. The pattern of a mosaic plot depends on the order in which the categorical variables are drawn. The categories of the first variable divide up the horizontal axis into columns whose widths

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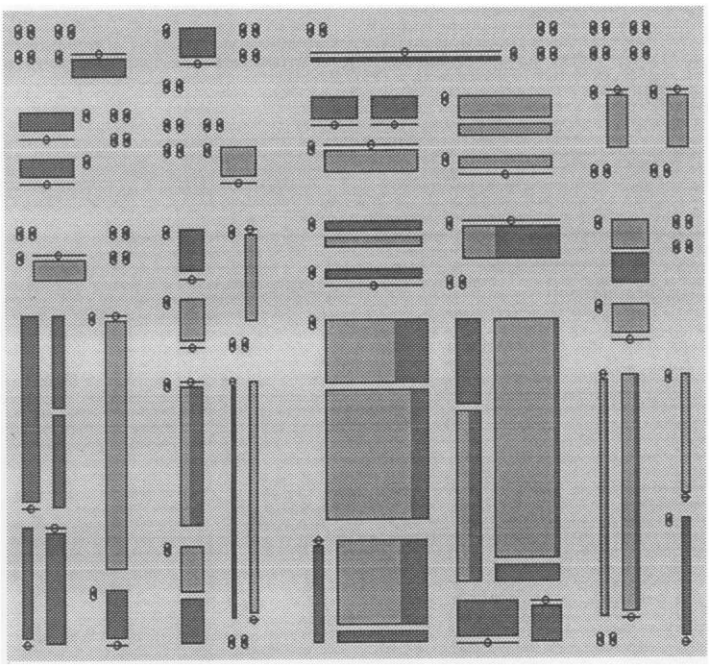

Figure 6. Mosaic Plot of the Students Who Answered Different Combinations of the Eight Questions.

are proportional to their numbers. The categories of the second variable divide up the vertical axis similarly but conditional on the first variable. The third variable further divides up the horizontal axis within each class of the first variable and so on.

Tricky problems arise in ensuring that the resulting blocks have equal areas for equal cell numbers. This problem can arise even with simple bar charts and histograms but is less likely to be of consequence or to be noticed. Because MANET's mosaic plots are fully interactive, linked, and can be interrogated there are many ways of ensuring that false impressions are avoided. Figure 6 shows the mosaic plot of which questions were answered by the 110 students. Each question was coded "yes" or "no." Some students did not answer as many as five questions and some students answered more than five. The highlighted students (the lighter grey) are those who passed.

1.5 STANDARD FEATURES OF MANET

MANET includes several other standard plotting features, but usually provides more information or additional flexibility:

- All plot classes can be interrogated by option-clicking with the mouse to show boundaries or class names (for bar charts), class numbers, and number currently selected.
- All plots may be rescaled or reformatted using a dialog box. The number of classes in a histogram may be changed interactively from the keyboard using the arrow keys as was implemented in REGARD. (Current work includes improving

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this by adding scale controls to change class width rather than the number of classes and to allow moving the histogram's origin). MANET includes the same scatterplot features as in REGARD for changing point size interactively (again with the arrow keys) and for displaying overlapping points by adding brightness values.

- When a plot scale is selected that results in some cases not being displayed (for instance, if a histogram for a variable whose range was from 0 to 50 was plotted from 20 to 40), then a red frame appears inside the plot to alert the analyst to the existence of points outside the current range. The same applies to scatterplots. Rescaling via dialogue boxes to set the parameters is common to many packages but few remind the user that values are missing.
- Often it is useful to give plots a common scale. Data Desk has a particularly clever feature for this whereby the scale for one histogram (and its window size) can be set equal to another's by dragging the latter's special icon box from its window-title bar into the target window. Unfortunately, this can only be done one window at a time and so can be clumsy (for instance, when we want to scale the histograms for the eight questions equally). MANET provides a menu option to choose windows from a scrolling list, all of which are then scaled to the smallest scale which contains them all.
- In all plots it is possible to ignore missing values using a command from the plot dialog window.

MANET does not provide any analytic features, as it is assumed these can be provided by other packages. In particular there are no models for predicting missing values included. MANET is for displaying data, not for modeling data.

1.6 Additional Features and Future Developments

MANET implements brushing as well as basic linking. This applies to all plots, including boxplots, where the highlighting scheme introduced in REGARD is used of a standard boxplot for the highlighted cases superimposed on a nonstandard boxplot for all cases. It is intended to experiment with ideas of generalized brushing (Craig, Haslett, Unwin, and Wills 1989) in the near future. A version was implemented in SPIDER, a forerunner of REGARD (Wills, Haslett, Unwin, and Craig 1989), which showed considerable promise.

There is often a need to record groups in the data on the fly and MANET provides a test version of this feature as well as group color-coding tools. To be fully consistent, colors have not only been used for individual points in scatterplots but also for displays in which points are grouped (bar charts, histograms, mosaic plots). This is experimental work (think of how you would display a histogram bar in which 8 of 12 blues are highlighted, 4 of 10 reds and 2 of 4 cases which are in both the blue and red groups) that will be reported on in another article.

Much of the stimulation for the developments in MANET arose from work on surveys where nonresponse and refusal lead to major missing value problems. Another issue in surveys is the use of weighting to adjust results. Although weighting can be partly

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provided for in interactive graphics by using a linked histogram of the weights, it is more natural to try to display the adjusted data directly. Another feature in MANET implements weighted histograms and bar charts with linking and interrogation. In keeping with the aim of the package to deal with missings effectively the displays have been additionally extended to cope with missings in the weighting variable. Further developments must include extension of these ideas to scatterplots and some solution for boxplots.

1.7 SUMMARY

The potential of interactive graphics tools has barely been tapped. MANET has introduced variations of the standard plots to effectively take account of missing values, but the software is also proving to be a valuable tool for experimenting with new interactive ideas.

1.8 SOFTWARE

MANET is written in C++ and runs only on Macintosh computers. Contact the first author for further information.

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